

1. (6 pts) Give examples of three groups of order 10.
2. (4 pts) Let a , b , and c be elements of a group. Solve the equation $axb = c$ for x .
3. (6 pts) Explain the difference between the order of a group, G , and the order of an element, x , in the group.
4. (12 pts) What is the inverse of each element in the given group.
 - a) Inverse of 3 in $\mathbb{Z}_{20}$
 - b) Inverse of 3 in $U(20)$
 - c) Inverse of H in D_{20}
 - d) Inverse of a^7 in $\langle a \rangle$ if $|a| = 20$
5. (9 pts) Refer to the Cayley table to answer the questions about the group G .

	1	2	3	4	5	6	7	8
1	1	2	3	4	5	6	7	8
2	2	3	4	1	8	5	6	7
3	3	4	1	2	7	8	5	6
4	4	1	2	3	6	7	8	5
5	5	6	7	8	3	4	1	2
6	6	7	8	5	2	3	4	1
7	7	8	5	6	1	2	3	4
8	8	5	6	7	4	1	2	3

- a) Give an example of a subgroup of order 2 in G .
 - b) What is the centralizer of 3 in G ?
 - c) What is the inverse of 5?
6. (8 pts) Explain why the following subsets are **not** subgroups of the given group.
 - a) $U(8)$ is not a subgroup of $\mathbb{Z}_8$
 - b) The set of nonnegative integers is not a subgroup of $(\mathbb{Z}, +)$
 7. (9 pts) Consider the dihedral group D_{11} .
 - a) How many elements are in D_{11} ?
 - b) List all possible orders for elements in D_{11} ?
 - c) How many elements are there with each order?
 8. (6 pts) Draw the subgroup lattice for $\mathbb{Z}_{50}$.

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9. (8 pts) Prove that $U(7)$ is a **cyclic** group. (You may assume it is a group.)
10. (12 pts) Let G be a group and let H be any subgroup of G . For any x and y in G , define $x \sim y$ if and only if $xy^{-1} \in H$. Prove that $\sim$ is an equivalence relation on the set G .
11. (10 pts) Prove that every cyclic group must be abelian.
12. (10 pts) Let G be an abelian group. Let H be the set of all elements of G that satisfy the equation $x^3 = e$. Prove that H is a subgroup of G .